



ŘEDITELSTVÍ SILNIC A DÁLNIC ČR

**NDIC - DATEX II Measured Data
Publication - Weather Data**

Release 1.0rev0

Národní Dopravní Informační Centrum

2018-04-29

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Identifier and version:

uri
cz-ndic_d2-weather-data-v1.0

Format
1.0

Specification
1.0rev0

Updated
2018-04-29

Format type:

Publication
DATEX II v2.3/MeasuredDataPublication

Extends
—

Publisher and author:

Publisher
NDIC

Author
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This Specification defines data format, used for publishing information about data read from meteo-stations.

INTRODUCTION

This document shall be used by publisher and / or subscriber. Subscriber shall use it for format suitability evaluation and for possible implementation of received NDIC data import into own information system. Publisher shall use it as a specification of data to be published format and for possible implementation of data export from own information system into NDIC.

Here you can find:

- conceptual description of published information
- information about location referencing methods used
- W3C XML schema of the format
- data sample(s)

Following topics are out of scope of this document and if present, shall be considered as informative only:

- transfer protocol incl. access point, life cycle etc.
- related data formats
- typical data origin
- typical data use

1.1 General terms

Following terms are used in this document:

format

description of data structure for transferring some kind of information

protocol

description of data transfer process and rules

NDIC

national traffic information centre

DATEX II model

class model for describing information related to the road traffic. The model in form of UML is platform independent.

One of specific platforms for expressing the model is XML.

The model is extensive (it consists of few thousands of structural items representing term incl. definition) For reference data (UML class model in form of Enterprise Architect EAP file, XML schema etc.) see [\[d2ref\]](#). For additional information see [\[d2supp\]](#).

(DATEX II) profile

simplified class model, still sufficient for specific needs. Profile is created from model by selecting only specific parts.

(DATEX II) extension

extended DATEX II model, adding data structures missing in standard model but needed for specific needs.

B level extension

Extension of DATEX II model, which is backward compatible with the original DATEX II model, even though it contains additional structures. [16157-1]

C level extension

Extension of DATEX II model, which is not backward compatible with the original DATEX II model.

publication

an element of DATEX II model, serving as an envelope for transmitted content of certain kind. Every sent DATEX II message contains exactly one element of publication type. There are various types of publications, for example *SituationPublication* [16157-3], *ElaboratedDataPublication* [16157-5], *PredefinedLocationsPublication* [16157-3] etc.

(W3C) XML schema

language, which allows definition of rules for a XML document structure.

XSD

XML document containing XML schema. Sometimes it is synonym for XML schema itself.

uri

uniform resource identifier, text, used as unique identifier. In this document, every format has it's own uri.

CONCEPTS

Described format allows publishing weather data (humidity, pollution, precipitation, road surface conditions, temperature, visibility and wind) for set of predefined weather station sites.

2.1 Publication type *MeasuredDataPublication*

Format is based on DATEX II publication of type *MeasuredDataPublication* [16157-5] and is used to publish traffic data for a set of predefined measuring equipment.

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <d2LogicalModel modelBaseVersion="2" xmlns="http://datex2.eu/schema/2/2_0" xmlns:xsd=
   ↳ "http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
   ↳ instance">
3   <exchange>
4     <supplierIdentification>
5       <country>cz</country>
6       <nationalIdentifier>DRU</nationalIdentifier>
7     </supplierIdentification>
8   </exchange>
9   <payloadPublication lang="cs" xsi:type="MeasuredDataPublication">
10    <feedType>x-format:cz-ndic_d2-weather-data-v1.0</feedType>
11    <publicationTime>2018-03-21T15:59:20+02:00</publicationTime>
12    <publicationCreator>
13      <country>cz</country>
14      <nationalIdentifier>DRU</nationalIdentifier>
15    </publicationCreator>
```

Traffic data is provided in element *measurementSiteReference* for a particular measuring equipment defined in a static publication *MeasurementSiteTablePublication* that is referenced via its unique id and version in the element *measurementSiteTableReference*.

Element *measurementSiteReference* with a set of measurements for a particular measuring site should appear in *siteMeasurements* multiple times (each time for different weather station).

2.2 Predefined measurement sites

The *measuring equipment* capability and location does not change. The measurement station is therefore, in this profile, described by a reference (id) to a predefined measurement site table publication defined in different document. The reference points to a class instance *MeasurementSiteRecord* from package *MeasurementSiteTablePublication*. Therefore, the location and capabilities of the equipment is not described directly in the file.

The equipment information (location and capabilities) is determined by reference to a set of predefined equipment. To look up given equipment, it is necessary to obtain publication *MeasurementSiteTablePublication*, related to this source of information and inside find location of identical type (by *targetClass*, here *MeasurementSiteRecord*), with the same identifier and version located within the same measurement table.

Fully qualified identifier of the measurement site consists of:

- measurement site table id and version and by
- measurement site record id and version.

```

16  <measurementSiteTableReference id="CZ_DRU_MST_WEA" targetClass=
17  ↪ "MeasurementSiteTable" version="1"/>
18  <headerInformation>
19    <confidentiality>noRestriction</confidentiality>
20    <informationStatus>real</informationStatus>
21  </headerInformation>
22  <siteMeasurements>
23  ↪ <measurementSiteReference id="12" targetClass="MeasurementSiteRecord" version="1
    <measurementTimeDefault>2018-03-21T15:59:20+02:00</measurementTimeDefault>

```

2.3 Weather station capabilities

Each weather station, in this profile, is capable of different set of measurements:

- humidity,
- pollution,
- precipitation,
- road surface conditions,
- temperature,
- visibility and
- wind.

The full list of possible measurements given by the weather station is called a capability and set in the *MeasurementSiteTablePublication* for each weather station verbatim as a number of *measurementSpecificCharacteristics* identified within the record by an index. To refer to that particular capability (i.e. precipitation) from this publication the capability index is used.

```

25  <measuredValue index="1">
26    <measuredValue>
27      <basicData xsi:type="PrecipitationInformation">
28        <precipitationDetail>
29          <precipitationType>rain</precipitationType>
30          <precipitationIntensity>
31            <millimetresPerHourIntensity>10</millimetresPerHourIntensity>
32          </precipitationIntensity>
33        </precipitationDetail>

```

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```

34     </basicData>
35     </measuredValue>
36 </measuredValue>
37 <!--road_status -->
38 <measuredValue index="2">
39     <measuredValue>
40         <basicData xsi:type="RoadSurfaceConditionInformation">
41             <weatherRelatedRoadConditionType>surfaceWater</
↪ weatherRelatedRoadConditionType>
42             <roadSurfaceConditionMeasurements>
43                 <waterFilmThickness>
44                     <floatingPointMetreDistance>0.01</floatingPointMetreDistance>
45                 </waterFilmThickness>
46             </roadSurfaceConditionMeasurements>
47         </basicData>
48     </measuredValue>
49 </measuredValue>

```

2.4 Publishing only currently available data

Format provides information about capabilities with known status. Either value is known (could be zero) or the capability is in faulty state (i.e. unknown). Only situation when respective capability *measuredValue* with given index value is not present is when the value could not be calculated (i.e. for rain intensity when there is no rain).

TIME VALIDITY

The time of validity for all measured data at one measurement site is given by the mandatory element *measurementTimeDefault* nested in *measurementSiteReference*. So each measurement site has to have its own validity time set up. Validity time means end of the measurement period, when data is calculated and provided.

No further specification of validity is used.

```
10 <feedType>x-format:cz-ndic_d2-weather-data-v1.0</feedType>
11 <publicationTime>2018-03-21T15:59:20+02:00</publicationTime>
12 <publicationCreator>
13   <country>cz</country>
14   <nationalIdentifier>DRU</nationalIdentifier>
15 </publicationCreator>
16 <measurementSiteTableReference id="CZ_DRU_MST_WEA" targetClass=
17 ↪ "MeasurementSiteTable" version="1"/>
18 <headerInformation>
19   <confidentiality>noRestriction</confidentiality>
20   <informationStatus>real</informationStatus>
21 </headerInformation>
22 <siteMeasurements>
23 ↪   <measurementSiteReference id="12" targetClass="MeasurementSiteRecord" version="1
   <measurementTimeDefault>2018-03-21T15:59:20+02:00</measurementTimeDefault>
```

LOCATION REFERENCING

The format does not describe locations, they are expected as a part of equipment specification in publication *MeasurementSiteTablePublication* [16157-5].

Location referencing methods are out of scope of this document and are described elsewhere.

APPENDIX A: MESSAGE EXAMPLES

5.1 meteo-data.xml - weather data measured at one equipment

Publication contains one record for one equipment capable to measure several weather related characteristics.

Equipment details (capabilities and location) are defined in different publication of type *MeasurementSiteTablePublication*, which is not described here in detail.

The equipment, is identified by a measurement site table reference *id*="CZ_DRU_MST_WEA" (with version), and by measurement site reference *id*="12", (with version).

```
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel modelBaseVersion="2" xmlns="http://datex2.eu/schema/2/2_0" xmlns:xsd=
  ↪ "http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
  ↪ instance">
  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>DRU</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication lang="cs" xsi:type="MeasuredDataPublication">
    <feedType>x-format:cz-ndic_d2-weather-data-v1.0</feedType>
    <publicationTime>2018-03-21T15:59:20+02:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>DRU</nationalIdentifier>
    </publicationCreator>
    <measurementSiteTableReference id="CZ_DRU_MST_WEA" targetClass=
  ↪ "MeasurementSiteTable" version="1"/>
    <headerInformation>
      <confidentiality>noRestriction</confidentiality>
      <informationStatus>real</informationStatus>
    </headerInformation>
    <siteMeasurements>
      <measurementSiteReference id="12" targetClass="MeasurementSiteRecord" version="1
  ↪ "/>
      <measurementTimeDefault>2018-03-21T15:59:20+02:00</measurementTimeDefault>
      <!--precip_type + precipitation -->
      <measuredValue index="1">
        <measuredValue>
          <basicData xsi:type="PrecipitationInformation">
            <precipitationDetail>
              <precipitationType>rain</precipitationType>
              <precipitationIntensity>
                <millimetresPerHourIntensity>10</millimetresPerHourIntensity>
```

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```

        </precipitationIntensity>
      </precipitationDetail>
    </basicData>
  </measuredValue>
</measuredValue>
<!--road_status -->
<measuredValue index="2">
  <measuredValue>
    <basicData xsi:type="RoadSurfaceConditionInformation">
      <weatherRelatedRoadConditionType>surfaceWater</
↪weatherRelatedRoadConditionType>
      <roadSurfaceConditionMeasurements>
        <waterFilmThickness>
          <floatingPointMetreDistance>0.01</floatingPointMetreDistance>
        </waterFilmThickness>
      </roadSurfaceConditionMeasurements>
    </basicData>
  </measuredValue>
</measuredValue>
<!--road_temp 1 + freez_temp-->
<measuredValue index="3">
  <measuredValue>
    <basicData xsi:type="RoadSurfaceConditionInformation">
      <roadSurfaceConditionMeasurements>
        <roadSurfaceTemperature>
          <temperature>8.7</temperature>
        </roadSurfaceTemperature>
        <protectionTemperature>
          <temperature>0</temperature>
        </protectionTemperature>
      </roadSurfaceConditionMeasurements>
    </basicData>
  </measuredValue>
</measuredValue>
<!--road_temp 2 + freez_temp-->
<measuredValue index="4">
  <measuredValue>
    <measurementEquipmentFault>
      <faultLastUpdateTime>2018-03-21T15:59:20+02:00</faultLastUpdateTime>
      <measurementEquipmentFault>noDataValuesAvailable</
↪measurementEquipmentFault>
    </measurementEquipmentFault>
  </measuredValue>
</measuredValue>
<!--air_temp + dewpoint -->
<measuredValue index="6">
  <measuredValue>
    <basicData xsi:type="TemperatureInformation">
      <temperature>
        <airTemperature>
          <temperature>8.1</temperature>
        </airTemperature>
        <dewPointTemperature>
          <temperature>0.1</temperature>
        </dewPointTemperature>
      </temperature>
    </basicData>
  </measuredValue>
</measuredValue>

```

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```

    </basicData>
  </measuredValue>
</measuredValue>
<!--humidity -->
<measuredValue index="7">
  <measuredValue>
    <basicData xsi:type="HumidityInformation">
      <humidity>
        <relativeHumidity>
          <percentage>57</percentage>
        </relativeHumidity>
      </humidity>
    </basicData>
  </measuredValue>
</measuredValue>
<!--wind_dir + wind_sp-->
<measuredValue index="8">
  <measuredValue>
    <basicData xsi:type="WindInformation">
      <wind>
        <windSpeed>
          <speed>1.7</speed>
        </windSpeed>
        <windDirectionBearing>
          <directionBearing>182</directionBearing>
        </windDirectionBearing>
      </wind>
    </basicData>
  </measuredValue>
</measuredValue>
<!-- visibility -->
<measuredValue index="9">
  <measuredValue>
    <basicData xsi:type="VisibilityInformation">
      <visibility>
        <minimumVisibilityDistance>
          <integerMetreDistance>2000</integerMetreDistance>
        </minimumVisibilityDistance>
      </visibility>
    </basicData>
  </measuredValue>
</measuredValue>
</siteMeasurements>
</payloadPublication>
</d2LogicalModel>

```

5.1.1 Root element and publication envelope

Root element *d2LogicalModel* declares, that this message is using DATEX II.

Element *exchange* contains provider identifier.

Element *payloadPublication* is an envelope for actual information payload containing general information, measurement site table reference *measurementSiteTableReference* and one element *siteMeasurements*.

Attribute *xsi:type="MeasuredDataPublication"* in element *payloadPublication* determines, that this envelope is of type Measured data publication.

Element *feedType* states publication schema by its uri.

Element *publicationTime* gives publication time for the whole publication which might be the time, for which were all traffic status values calculated (start of validity).

Element *publicationCreator* identifies creator of the publication.

5.1.2 Description of weather data at one place

The element *siteMeasurements* contains one or more measurements referenced to a particular road equipment *measurementSiteReference*. The *measurementSiteReference* is identified by id and version.

measurementTimeDefault element gives time of validity for the set of measurement within one measurement site. It is also the time, when measured weather data were created (i.e. the end of the measurement period).

Following element *measuredValue* can be repeated and describes a measured weather data for a particular measurement site.

The equipment (weather station) has 9 capabilities defined (defined in *MeasurementSiteTablePublication*). They are:

- precipitation,
- road surface conditions (3x),
- temperature,
- humidity,
- visibility and
- wind.

The road surface conditions is there 3 times (indexes 2-4). First one defines overall characteristics and is used to convey road status. Remaining two are road lane related measurements (one for each lane) the latter form the two is in faulty state *measurementEquipmentFault*.

Nested element *basicData* contains the actual measured value of the given type.

Time validity

Start of validity for measured data is stated by mandatory element *measurementTimeDefault*.

APPENDIX B: W3C XML SCHEMA

XML structure of a transmitted message is defined by following W3C XML schema:

6.1 *schema.xsd*

```

1 <?xml version="1.0" encoding="utf-8" standalone="no"?>
2 <xs:schema elementFormDefault="qualified" attributeFormDefault="unqualified"
  ↳ xmlns:D2LogicalModel="http://datex2.eu/schema/2/2_0" version="2.3" targetNamespace=
  ↳ "http://datex2.eu/schema/2/2_0" xmlns:xs="http://www.w3.org/2001/XMLSchema">
3   <xs:complexType name="_ExtensionType">
4     <xs:sequence>
5       <xs:any namespace="##any" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
6     </xs:sequence>
7   </xs:complexType>
8   <xs:complexType name="_MeasurementSiteRecordVersionedReference">
9     <xs:complexContent>
10      <xs:extension base="D2LogicalModel:VersionedReference">
11        <xs:attribute name="targetClass" use="required" fixed="MeasurementSiteRecord"
  ↳ />
12      </xs:extension>
13    </xs:complexContent>
14  </xs:complexType>
15  <xs:complexType name="_MeasurementSiteTableVersionedReference">
16    <xs:complexContent>
17      <xs:extension base="D2LogicalModel:VersionedReference">
18        <xs:attribute name="targetClass" use="required" fixed="MeasurementSiteTable" /
  ↳ >
19      </xs:extension>
20    </xs:complexContent>
21  </xs:complexType>
22  <xs:complexType name="_SiteMeasurementsIndexMeasuredValue">
23    <xs:sequence>
24      <xs:element name="measuredValue" type="D2LogicalModel:MeasuredValue" minOccurs=
  ↳ "1" maxOccurs="1" />
25    </xs:sequence>
26    <xs:attribute name="index" type="xs:int" use="required" />
27  </xs:complexType>
28  <xs:simpleType name="AngleInDegrees">
29    <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
30  </xs:simpleType>
31  <xs:complexType name="ApplicationRateValue">
32    <xs:complexContent>
33      <xs:extension base="D2LogicalModel:DataValue">

```

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```

34     <xs:sequence>
35         <xs:element name="applicationRate" type=
36         ↪ "D2LogicalModel:IntensityKilogramsPerSquareMetre" minOccurs="1" maxOccurs="1" />
37         <xs:element name="applicationRateValueExtension" type="D2LogicalModel:_
38         ↪ ExtensionType" minOccurs="0" />
39     </xs:sequence>
40 </xs:extension>
41 </xs:complexType>
42 <xs:simpleType name="AreaOfInterestEnum">
43     <xs:restriction base="xs:string">
44         <xs:enumeration value="continentWide" />
45         <xs:enumeration value="national" />
46         <xs:enumeration value="neighbouringCountries" />
47         <xs:enumeration value="notSpecified" />
48         <xs:enumeration value="regional" />
49     </xs:restriction>
50 </xs:simpleType>
51 <xs:complexType name="BasicData" abstract="true">
52     <xs:sequence>
53         <xs:element name="measurementOrCalculationPeriod" type="D2LogicalModel:Seconds" ↪
54         ↪ minOccurs="0" maxOccurs="1" />
55         <xs:element name="measurementOrCalculationTime" type="D2LogicalModel:DateTime" ↪
56         ↪ minOccurs="0" maxOccurs="1" />
57         <xs:element name="basicDataExtension" type="D2LogicalModel:_ExtensionType" ↪
58         ↪ minOccurs="0" />
59     </xs:sequence>
60     <xs:attribute name="measurementOrCalculatedTimePrecision" type=
61     ↪ "D2LogicalModel:TimePrecisionEnum" use="optional" />
62 </xs:complexType>
63 <xs:simpleType name="Boolean">
64     <xs:restriction base="xs:boolean" />
65 </xs:simpleType>
66 <xs:simpleType name="ComputationMethodEnum">
67     <xs:restriction base="xs:string">
68         <xs:enumeration value="arithmeticAverageOfSamplesBasedOnAFixedNumberOfSamples" /
69         ↪ >
70         <xs:enumeration value="arithmeticAverageOfSamplesInATimePeriod" />
71         <xs:enumeration value="harmonicAverageOfSamplesInATimePeriod" />
72         <xs:enumeration value="medianOfSamplesInATimePeriod" />
73         <xs:enumeration value="movingAverageOfSamples" />
74     </xs:restriction>
75 </xs:simpleType>
76 <xs:simpleType name="ConcentrationKilogramsPerCubicMetre">
77     <xs:restriction base="D2LogicalModel:Float" />
78 </xs:simpleType>
79 <xs:simpleType name="ConcentrationMicrogramsPerCubicMetre">
80     <xs:restriction base="D2LogicalModel:Float" />
81 </xs:simpleType>
82 <xs:simpleType name="ConfidentialityValueEnum">
    <xs:restriction base="xs:string">
        <xs:enumeration value="internalUse" />
        <xs:enumeration value="noRestriction" />
        <xs:enumeration value="restrictedToAuthorities" />
        <xs:enumeration value="restrictedToAuthoritiesAndTrafficOperators" />
        <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndPublishers" />
    </xs:restriction>

```

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```

83     <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndVms" />
84   </xs:restriction>
85 </xs:simpleType>
86 <xs:simpleType name="CountryEnum">
87   <xs:restriction base="xs:string">
88     <xs:enumeration value="at" />
89     <xs:enumeration value="be" />
90     <xs:enumeration value="bg" />
91     <xs:enumeration value="ch" />
92     <xs:enumeration value="cs" />
93     <xs:enumeration value="cy" />
94     <xs:enumeration value="cz" />
95     <xs:enumeration value="de" />
96     <xs:enumeration value="dk" />
97     <xs:enumeration value="ee" />
98     <xs:enumeration value="es" />
99     <xs:enumeration value="fi" />
100    <xs:enumeration value="fo" />
101    <xs:enumeration value="fr" />
102    <xs:enumeration value="gb" />
103    <xs:enumeration value="gg" />
104    <xs:enumeration value="gi" />
105    <xs:enumeration value="gr" />
106    <xs:enumeration value="hr" />
107    <xs:enumeration value="hu" />
108    <xs:enumeration value="ie" />
109    <xs:enumeration value="im" />
110    <xs:enumeration value="is" />
111    <xs:enumeration value="it" />
112    <xs:enumeration value="je" />
113    <xs:enumeration value="li" />
114    <xs:enumeration value="lt" />
115    <xs:enumeration value="lu" />
116    <xs:enumeration value="lv" />
117    <xs:enumeration value="ma" />
118    <xs:enumeration value="mc" />
119    <xs:enumeration value="mk" />
120    <xs:enumeration value="mt" />
121    <xs:enumeration value="nl" />
122    <xs:enumeration value="no" />
123    <xs:enumeration value="pl" />
124    <xs:enumeration value="pt" />
125    <xs:enumeration value="ro" />
126    <xs:enumeration value="se" />
127    <xs:enumeration value="si" />
128    <xs:enumeration value="sk" />
129    <xs:enumeration value="sm" />
130    <xs:enumeration value="tr" />
131    <xs:enumeration value="va" />
132    <xs:enumeration value="other" />
133   </xs:restriction>
134 </xs:simpleType>
135 <xs:element name="d2LogicalModel" type="D2LogicalModel:D2LogicalModel" />
136 <xs:complexType name="D2LogicalModel">
137   <xs:sequence>
138     <xs:element name="exchange" type="D2LogicalModel:Exchange" />

```

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```

139     <xs:element name="payloadPublication" type="D2LogicalModel:PayloadPublication"
↪ minOccurs="0" />
140     <xs:element name="d2LogicalModelExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
141   </xs:sequence>
142   <xs:attribute name="modelBaseVersion" use="required" fixed="2" />
143 </xs:complexType>
144 <xs:complexType name="DataValue" abstract="true">
145   <xs:sequence>
146     <xs:element name="dataError" type="D2LogicalModel:Boolean" minOccurs="0"
↪ maxOccurs="1" />
147     <xs:element name="reasonForDataError" type="D2LogicalModel:MultilingualString"
↪ minOccurs="0" maxOccurs="1" />
148     <xs:element name="dataValueExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
149   </xs:sequence>
150   <xs:attribute name="accuracy" type="D2LogicalModel:Percentage" use="optional" />
151   <xs:attribute name="computationalMethod" type=
↪ "D2LogicalModel:ComputationMethodEnum" use="optional" />
152   <xs:attribute name="numberOfIncompleteInputs" type=
↪ "D2LogicalModel:NonNegativeInteger" use="optional" />
153   <xs:attribute name="numberOfInputValuesUsed" type=
↪ "D2LogicalModel:NonNegativeInteger" use="optional" />
154   <xs:attribute name="smoothingFactor" type="D2LogicalModel:Float" use="optional" />
155   <xs:attribute name="standardDeviation" type="D2LogicalModel:Float" use="optional"
↪ />
156   <xs:attribute name="supplierCalculatedDataQuality" type="D2LogicalModel:Percentage
↪ " use="optional" />
157 </xs:complexType>
158 <xs:simpleType name="DateTime">
159   <xs:restriction base="xs:dateTime" />
160 </xs:simpleType>
161 <xs:complexType name="DirectionBearingValue">
162   <xs:complexContent>
163     <xs:extension base="D2LogicalModel:DataValue">
164       <xs:sequence>
165         <xs:element name="directionBearing" type="D2LogicalModel:AngleInDegrees"
↪ minOccurs="1" maxOccurs="1" />
166         <xs:element name="directionBearingValueExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
167       </xs:sequence>
168     </xs:extension>
169   </xs:complexContent>
170 </xs:complexType>
171 <xs:simpleType name="DirectionCompassEnum">
172   <xs:restriction base="xs:string">
173     <xs:enumeration value="east" />
174     <xs:enumeration value="eastNorthEast" />
175     <xs:enumeration value="eastSouthEast" />
176     <xs:enumeration value="north" />
177     <xs:enumeration value="northEast" />
178     <xs:enumeration value="northNorthEast" />
179     <xs:enumeration value="northNorthWest" />
180     <xs:enumeration value="northWest" />
181     <xs:enumeration value="south" />
182     <xs:enumeration value="southEast" />

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183     <xs:enumeration value="southSouthEast" />
184     <xs:enumeration value="southSouthWest" />
185     <xs:enumeration value="southWest" />
186     <xs:enumeration value="west" />
187     <xs:enumeration value="westNorthWest" />
188     <xs:enumeration value="westSouthWest" />
189   </xs:restriction>
190 </xs:simpleType>
191 <xs:complexType name="DirectionCompassValue">
192   <xs:complexContent>
193     <xs:extension base="D2LogicalModel:DataValue">
194       <xs:sequence>
195         <xs:element name="directionCompass" type=
196 ↪ "D2LogicalModel:DirectionCompassEnum" minOccurs="1" maxOccurs="1" />
197         <xs:element name="directionCompassValueExtension" type="D2LogicalModel:_
198 ↪ ExtensionType" minOccurs="0" />
199       </xs:sequence>
200     </xs:extension>
201   </xs:complexContent>
202 </xs:complexType>
203 <xs:complexType name="ElaboratedDataFault">
204   <xs:complexContent>
205     <xs:extension base="D2LogicalModel:Fault">
206       <xs:sequence>
207         <xs:element name="elaboratedDataFault" type=
208 ↪ "D2LogicalModel:ElaboratedDataFaultEnum" minOccurs="1" maxOccurs="1" />
209         <xs:element name="elaboratedDataFaultExtension" type="D2LogicalModel:_
210 ↪ ExtensionType" minOccurs="0" />
211       </xs:sequence>
212     </xs:extension>
213   </xs:complexContent>
214 </xs:complexType>
215 <xs:simpleType name="ElaboratedDataFaultEnum">
216   <xs:restriction base="xs:string">
217     <xs:enumeration value="intermittentDataValues" />
218     <xs:enumeration value="noDataValuesAvailable" />
219     <xs:enumeration value="spuriousUnreliableDataValues" />
220     <xs:enumeration value="unspecifiedOrUnknownFault" />
221     <xs:enumeration value="other" />
222   </xs:restriction>
223 </xs:simpleType>
224 <xs:complexType name="Exchange">
225   <xs:sequence>
226     <xs:element name="supplierIdentification" type=
227 ↪ "D2LogicalModel:InternationalIdentifier" />
228     <xs:element name="exchangeExtension" type="D2LogicalModel:_ExtensionType"
229 ↪ minOccurs="0" />
230   </xs:sequence>
231 </xs:complexType>
232 <xs:complexType name="Fault">
233   <xs:sequence>
234     <xs:element name="faultIdentifier" type="D2LogicalModel:String" minOccurs="0"
235 ↪ maxOccurs="1" />
236     <xs:element name="faultDescription" type="D2LogicalModel:String" minOccurs="0"
237 ↪ maxOccurs="1" />
238     <xs:element name="faultCreationTime" type="D2LogicalModel:DateTime" minOccurs="0

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231   <xs:element name="faultLastUpdateTime" type="D2LogicalModel:DateTime" minOccurs=
232   <xs:element name="faultSeverity" type="D2LogicalModel:FaultSeverityEnum"
233   <xs:element name="faultExtension" type="D2LogicalModel:_ExtensionType"
234   </xs:sequence>
235   </xs:complexType>
236   <xs:simpleType name="FaultSeverityEnum">
237     <xs:restriction base="xs:string">
238       <xs:enumeration value="low" />
239       <xs:enumeration value="medium" />
240       <xs:enumeration value="high" />
241       <xs:enumeration value="unknown" />
242     </xs:restriction>
243   </xs:simpleType>
244   <xs:simpleType name="Float">
245     <xs:restriction base="xs:float" />
246   </xs:simpleType>
247   <xs:complexType name="FloatingPointMetreDistanceValue">
248     <xs:complexContent>
249       <xs:extension base="D2LogicalModel:DataValue">
250         <xs:sequence>
251           <xs:element name="floatingPointMetreDistance" type=
252           <xs:element name="floatingPointMetreDistanceValueExtension" type=
253           </xs:sequence>
254         </xs:extension>
255       </xs:complexContent>
256     </xs:complexType>
257   <xs:complexType name="HeaderInformation">
258     <xs:sequence>
259       <xs:element name="areaOfInterest" type="D2LogicalModel:AreaOfInterestEnum"
260       <xs:element name="confidentiality" type="D2LogicalModel:ConfidentialityValueEnum"
261       <xs:element name="informationStatus" type="D2LogicalModel:InformationStatusEnum"
262       <xs:element name="urgency" type="D2LogicalModel:UrgencyEnum" minOccurs="0"
263       <xs:element name="headerInformationExtension" type="D2LogicalModel:_
264       </xs:sequence>
265     </xs:complexType>
266   <xs:complexType name="Humidity">
267     <xs:sequence>
268       <xs:element name="relativeHumidity" type="D2LogicalModel:PercentageValue" />
269       <xs:element name="humidityExtension" type="D2LogicalModel:_ExtensionType"
270     </xs:sequence>
271   </xs:complexType>
272   <xs:complexType name="HumidityInformation">
273     <xs:complexContent>
274     <xs:extension base="D2LogicalModel:WeatherData">

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275     <xs:sequence>
276       <xs:element name="humidity" type="D2LogicalModel:Humidity" />
277       <xs:element name="humidityInformationExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
278     </xs:sequence>
279   </xs:extension>
280 </xs:complexContent>
281 </xs:complexType>
282 <xs:simpleType name="InformationStatusEnum">
283   <xs:restriction base="xs:string">
284     <xs:enumeration value="real" />
285     <xs:enumeration value="securityExercise" />
286     <xs:enumeration value="technicalExercise" />
287     <xs:enumeration value="test" />
288   </xs:restriction>
289 </xs:simpleType>
290 <xs:simpleType name="Integer">
291   <xs:restriction base="xs:integer" />
292 </xs:simpleType>
293 <xs:complexType name="IntegerMetreDistanceValue">
294   <xs:complexContent>
295     <xs:extension base="D2LogicalModel:DataValue">
296       <xs:sequence>
297         <xs:element name="integerMetreDistance" type=
↪"D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="1" maxOccurs="1" />
298         <xs:element name="integerMetreDistanceValueExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
299       </xs:sequence>
300     </xs:extension>
301   </xs:complexContent>
302 </xs:complexType>
303 <xs:simpleType name="IntensityKilogramsPerSquareMetre">
304   <xs:restriction base="D2LogicalModel:Float" />
305 </xs:simpleType>
306 <xs:simpleType name="IntensityMillimetresPerHour">
307   <xs:restriction base="D2LogicalModel:Float" />
308 </xs:simpleType>
309 <xs:complexType name="InternationalIdentifier">
310   <xs:sequence>
311     <xs:element name="country" type="D2LogicalModel:CountryEnum" minOccurs="1"
↪maxOccurs="1" />
312     <xs:element name="nationalIdentifier" type="D2LogicalModel:String" minOccurs="1"
↪maxOccurs="1" />
313     <xs:element name="internationalIdentifierExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
314   </xs:sequence>
315 </xs:complexType>
316 <xs:complexType name="KilogramsConcentrationValue">
317   <xs:complexContent>
318     <xs:extension base="D2LogicalModel:DataValue">
319       <xs:sequence>
320         <xs:element name="kilogramsConcentration" type=
↪"D2LogicalModel:ConcentrationKilogramsPerCubicMetre" minOccurs="1" maxOccurs="1" />
321         <xs:element name="kilogramsConcentrationValueExtension" type=
↪"D2LogicalModel:_ExtensionType" minOccurs="0" />
322       </xs:sequence>

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323     </xs:extension>
324   </xs:complexContent>
325 </xs:complexType>
326 <xs:simpleType name="KilometresPerHour">
327   <xs:restriction base="D2LogicalModel:Float" />
328 </xs:simpleType>
329 <xs:simpleType name="LaneEnum">
330   <xs:restriction base="xs:string">
331     <xs:enumeration value="allLanesCompleteCarriageway" />
332     <xs:enumeration value="busLane" />
333     <xs:enumeration value="busStop" />
334     <xs:enumeration value="carPoolLane" />
335     <xs:enumeration value="centralReservation" />
336     <xs:enumeration value="crawlerLane" />
337     <xs:enumeration value="emergencyLane" />
338     <xs:enumeration value="escapeLane" />
339     <xs:enumeration value="expressLane" />
340     <xs:enumeration value="hardShoulder" />
341     <xs:enumeration value="heavyVehicleLane" />
342     <xs:enumeration value="lane1" />
343     <xs:enumeration value="lane2" />
344     <xs:enumeration value="lane3" />
345     <xs:enumeration value="lane4" />
346     <xs:enumeration value="lane5" />
347     <xs:enumeration value="lane6" />
348     <xs:enumeration value="lane7" />
349     <xs:enumeration value="lane8" />
350     <xs:enumeration value="lane9" />
351     <xs:enumeration value="layBy" />
352     <xs:enumeration value="leftHandTurningLane" />
353     <xs:enumeration value="leftLane" />
354     <xs:enumeration value="localTrafficLane" />
355     <xs:enumeration value="middleLane" />
356     <xs:enumeration value="opposingLanes" />
357     <xs:enumeration value="overtakingLane" />
358     <xs:enumeration value="rightHandTurningLane" />
359     <xs:enumeration value="rightLane" />
360     <xs:enumeration value="rushHourLane" />
361     <xs:enumeration value="setDownArea" />
362     <xs:enumeration value="slowVehicleLane" />
363     <xs:enumeration value="throughTrafficLane" />
364     <xs:enumeration value="tidalFlowLane" />
365     <xs:enumeration value="turningLane" />
366     <xs:enumeration value="verge" />
367   </xs:restriction>
368 </xs:simpleType>
369 <xs:simpleType name="Language">
370   <xs:restriction base="xs:language" />
371 </xs:simpleType>
372 <xs:complexType name="LocationCharacteristicsOverride">
373   <xs:sequence>
374     <xs:element name="measurementLanesOverride" type="D2LogicalModel:LaneEnum"
375     ↪ minOccurs="0" maxOccurs="1" />
376     <xs:element name="reversedFlow" type="D2LogicalModel:Boolean" minOccurs="0"
377     ↪ maxOccurs="1" />
378     <xs:element name="locationCharacteristicsOverrideExtension" type="

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377   <xs:sequence>
378 </xs:complexType>
379 <xs:complexType name="MeasuredDataPublication">
380   <xs:complexContent>
381     <xs:extension base="D2LogicalModel:PayloadPublication">
382       <xs:sequence>
383         <xs:element name="measurementSiteTableReference" type="D2LogicalModel:_
384   <MeasurementSiteTableVersionedReference" minOccurs="1" maxOccurs="1" />
385         <xs:element name="headerInformation" type="D2LogicalModel:HeaderInformation
386   <" />
387         <xs:element name="siteMeasurements" type="D2LogicalModel:SiteMeasurements"
388   <maxOccurs="unbounded" />
389         <xs:element name="measuredDataPublicationExtension" type="D2LogicalModel:_
390   <ExtensionType" minOccurs="0" />
391       </xs:sequence>
392     </xs:extension>
393   </xs:complexContent>
394 </xs:complexType>
395 <xs:complexType name="MeasuredValue">
396   <xs:sequence>
397     <xs:element name="measurementEquipmentTypeUsed" type=
398   <"D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
399     <xs:element name="locationCharacteristicsOverride" type=
400   <"D2LogicalModel:LocationCharacteristicsOverride" minOccurs="0" />
401     <xs:element name="measurementEquipmentFault" type=
402   <"D2LogicalModel:MeasurementEquipmentFault" minOccurs="0" maxOccurs="unbounded" />
403     <xs:element name="basicData" type="D2LogicalModel:BasicData" minOccurs="0" />
404     <xs:element name="measuredValueExtension" type="D2LogicalModel:_ExtensionType"
405   <minOccurs="0" />
406   </xs:sequence>
407 </xs:complexType>
408 <xs:complexType name="MeasurementEquipmentFault">
409   <xs:complexContent>
410     <xs:extension base="D2LogicalModel:Fault">
411       <xs:sequence>
412         <xs:element name="measurementEquipmentFault" type=
413   <"D2LogicalModel:MeasurementEquipmentFaultEnum" minOccurs="1" maxOccurs="1" />
414         <xs:element name="measurementEquipmentFaultExtension" type="D2LogicalModel:_
415   <ExtensionType" minOccurs="0" />
416       </xs:sequence>
417     </xs:extension>
418   </xs:complexContent>
419 </xs:complexType>
420 <xs:simpleType name="MeasurementEquipmentFaultEnum">
421   <xs:restriction base="xs:string">
    <xs:enumeration value="intermittentDataValues" />
    <xs:enumeration value="noDataValuesAvailable" />
    <xs:enumeration value="spuriousUnreliableDataValues" />
    <xs:enumeration value="unspecifiedOrUnknownFault" />
    <xs:enumeration value="other" />
  </xs:restriction>
</xs:simpleType>
<xs:simpleType name="MetresAsFloat">
  <xs:restriction base="D2LogicalModel:Float" />
</xs:simpleType>

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422 <xs:simpleType name="MetresAsNonNegativeInteger">
423   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
424 </xs:simpleType>
425 <xs:complexType name="MicrogramsConcentrationValue">
426   <xs:complexContent>
427     <xs:extension base="D2LogicalModel:DataValue">
428       <xs:sequence>
429         <xs:element name="microgramsConcentration" type=
430 ↪ "D2LogicalModel:ConcentrationMicrogramsPerCubicMetre" minOccurs="1" maxOccurs="1" />
431         <xs:element name="microgramsConcentrationValueExtension" type=
432 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
433       </xs:sequence>
434     </xs:extension>
435   </xs:complexContent>
436 </xs:complexType>
437 <xs:complexType name="MultilingualString">
438   <xs:sequence>
439     <xs:element name="values">
440       <xs:complexType>
441         <xs:sequence>
442           <xs:element name="value" type="D2LogicalModel:MultilingualStringValue"
443 ↪ maxOccurs="unbounded" />
444         </xs:sequence>
445       </xs:complexType>
446     </xs:element>
447   </xs:sequence>
448 </xs:complexType>
449 <xs:complexType name="MultilingualStringValue">
450   <xs:simpleContent>
451     <xs:extension base="D2LogicalModel:MultilingualStringValueType">
452       <xs:attribute name="lang" type="xs:language" />
453     </xs:extension>
454   </xs:simpleContent>
455 </xs:complexType>
456 <xs:simpleType name="MultilingualStringValueType">
457   <xs:restriction base="xs:string">
458     <xs:maxLength value="1024" />
459   </xs:restriction>
460 </xs:simpleType>
461 <xs:simpleType name="NonNegativeInteger">
462   <xs:restriction base="xs:nonNegativeInteger" />
463 </xs:simpleType>
464 <xs:complexType name="OccupancyChangeValue">
465   <xs:complexContent>
466     <xs:extension base="D2LogicalModel:DataValue">
467       <xs:sequence>
468         <xs:element name="occupancyChange" type="D2LogicalModel:Integer" minOccurs=
469 ↪ "1" maxOccurs="1" />
470         <xs:element name="occupancyChangeValueExtension" type="D2LogicalModel:_
471 ↪ ExtensionType" minOccurs="0" />
472       </xs:sequence>
473     </xs:extension>
474   </xs:complexContent>
475 </xs:complexType>
476 <xs:complexType name="PayloadPublication" abstract="true">
477   <xs:sequence>

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473     <xs:element name="feedType" type="D2LogicalModel:String" minOccurs="0"
↪maxOccurs="1" />
474     <xs:element name="publicationTime" type="D2LogicalModel:DateTime" minOccurs="1"
↪maxOccurs="1" />
475     <xs:element name="publicationCreator" type=
↪"D2LogicalModel:InternationalIdentifier" />
476     <xs:element name="payloadPublicationExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
477     </xs:sequence>
478     <xs:attribute name="lang" type="D2LogicalModel:Language" use="required" />
479 </xs:complexType>
480 <xs:simpleType name="Percentage">
481     <xs:restriction base="D2LogicalModel:Float" />
482 </xs:simpleType>
483 <xs:complexType name="PercentageValue">
484     <xs:complexContent>
485         <xs:extension base="D2LogicalModel:DataValue">
486             <xs:sequence>
487                 <xs:element name="percentage" type="D2LogicalModel:Percentage" minOccurs="1
↪" maxOccurs="1" />
488                 <xs:element name="percentageValueExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
489             </xs:sequence>
490         </xs:extension>
491     </xs:complexContent>
492 </xs:complexType>
493 <xs:simpleType name="PollutantTypeEnum">
494     <xs:restriction base="xs:string">
495         <xs:enumeration value="benzeneTolueneXylene" />
496         <xs:enumeration value="carbonMonoxide" />
497         <xs:enumeration value="lead" />
498         <xs:enumeration value="methane" />
499         <xs:enumeration value="nitricOxide" />
500         <xs:enumeration value="nitrogenDioxide" />
501         <xs:enumeration value="nitrogenMonoxide" />
502         <xs:enumeration value="nitrogenOxides" />
503         <xs:enumeration value="nonMethaneHydrocarbons" />
504         <xs:enumeration value="ozone" />
505         <xs:enumeration value="particulates10" />
506         <xs:enumeration value="polycyclicAromaticHydrocarbons" />
507         <xs:enumeration value="primaryParticulate" />
508         <xs:enumeration value="sulphurDioxide" />
509         <xs:enumeration value="totalHydrocarbons" />
510     </xs:restriction>
511 </xs:simpleType>
512 <xs:complexType name="Pollution">
513     <xs:sequence>
514         <xs:element name="pollutantType" type="D2LogicalModel:PollutantTypeEnum"
↪minOccurs="1" maxOccurs="1" />
515         <xs:element name="pollutantConcentration" type=
↪"D2LogicalModel:MicrogramsConcentrationValue" minOccurs="0" />
516         <xs:element name="pollutionExtension" type="D2LogicalModel:_ExtensionType"
↪minOccurs="0" />
517     </xs:sequence>
518 </xs:complexType>
519 <xs:complexType name="PollutionInformation">

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520     <xs:complexContent>
521       <xs:extension base="D2LogicalModel:WeatherData">
522         <xs:sequence>
523           <xs:element name="pollution" type="D2LogicalModel:Pollution" maxOccurs=
524             ↪ "unbounded" />
525           <xs:element name="pollutionInformationExtension" type="D2LogicalModel:_
526             ↪ ExtensionType" minOccurs="0" />
527         </xs:sequence>
528       </xs:extension>
529     </xs:complexContent>
530   </xs:complexType>
531   <xs:complexType name="PrecipitationDetail">
532     <xs:sequence>
533       <xs:element name="precipitationType" type="D2LogicalModel:PrecipitationTypeEnum
534         ↪ " minOccurs="0" maxOccurs="1" />
535       <xs:element name="precipitationIntensity" type=
536         ↪ "D2LogicalModel:PrecipitationIntensityValue" minOccurs="0" />
537       <xs:element name="depositionDepth" type=
538         ↪ "D2LogicalModel:FloatingPointMetreDistanceValue" minOccurs="0" />
539       <xs:element name="precipitationDetailExtension" type="D2LogicalModel:_
540         ↪ ExtensionType" minOccurs="0" />
541     </xs:sequence>
542   </xs:complexType>
543   <xs:complexType name="PrecipitationInformation">
544     <xs:complexContent>
545       <xs:extension base="D2LogicalModel:WeatherData">
546         <xs:sequence>
547           <xs:element name="noPrecipitation" type="D2LogicalModel:Boolean" minOccurs=
548             ↪ "0" maxOccurs="1" />
549           <xs:element name="precipitationDetail" type=
550             ↪ "D2LogicalModel:PrecipitationDetail" minOccurs="0" />
551           <xs:element name="precipitationInformationExtension" type="D2LogicalModel:_
552             ↪ ExtensionType" minOccurs="0" />
553         </xs:sequence>
554       </xs:extension>
555     </xs:complexContent>
556   </xs:complexType>
557   <xs:complexType name="PrecipitationIntensityValue">
558     <xs:complexContent>
559       <xs:extension base="D2LogicalModel:DataValue">
560         <xs:sequence>
561           <xs:element name="millimetresPerHourIntensity" type=
562             ↪ "D2LogicalModel:IntensityMillimetresPerHour" minOccurs="1" maxOccurs="1" />
563           <xs:element name="precipitationIntensityValueExtension" type=
564             ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
565         </xs:sequence>
566       </xs:extension>
567     </xs:complexContent>
568   </xs:complexType>
569   <xs:simpleType name="PrecipitationTypeEnum">
570     <xs:restriction base="xs:string">
571       <xs:enumeration value="drizzle" />
572       <xs:enumeration value="freezingRain" />
573       <xs:enumeration value="hail" />
574       <xs:enumeration value="rain" />
575       <xs:enumeration value="sleet" />

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565     <xs:enumeration value="snow" />
566   </xs:restriction>
567 </xs:simpleType>
568 <xs:complexType name="RoadSurfaceConditionInformation">
569   <xs:complexContent>
570     <xs:extension base="D2LogicalModel:WeatherData">
571       <xs:sequence>
572         <xs:element name="weatherRelatedRoadConditionType" type=
573 ↪ "D2LogicalModel:WeatherRelatedRoadConditionTypeEnum" minOccurs="0" maxOccurs=
574 ↪ "unbounded" />
575         <xs:element name="roadSurfaceConditionMeasurements" type=
576 ↪ "D2LogicalModel:RoadSurfaceConditionMeasurements" />
577         <xs:element name="roadSurfaceConditionInformationExtension" type=
578 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
579       </xs:sequence>
580     </xs:extension>
581   </xs:complexContent>
582 </xs:complexType>
583 <xs:complexType name="RoadSurfaceConditionMeasurements">
584   <xs:sequence>
585     <xs:element name="roadSurfaceTemperature" type="D2LogicalModel:TemperatureValue"
586 ↪ " minOccurs="0" />
587     <xs:element name="protectionTemperature" type="D2LogicalModel:TemperatureValue"
588 ↪ minOccurs="0" />
589     <xs:element name="deIcingApplicationRate" type=
590 ↪ "D2LogicalModel:ApplicationRateValue" minOccurs="0" />
591     <xs:element name="deIcingConcentration" type=
592 ↪ "D2LogicalModel:KilogramsConcentrationValue" minOccurs="0" />
593     <xs:element name="depthOfSnow" type=
594 ↪ "D2LogicalModel:FloatingPointMetreDistanceValue" minOccurs="0" />
595     <xs:element name="waterFilmThickness" type=
596 ↪ "D2LogicalModel:FloatingPointMetreDistanceValue" minOccurs="0" />
597     <xs:element name="roadSurfaceConditionMeasurementsExtension" type=
598 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
599   </xs:sequence>
600 </xs:complexType>
601 <xs:simpleType name="Seconds">
602   <xs:restriction base="D2LogicalModel:Float" />
603 </xs:simpleType>
604 <xs:complexType name="SiteMeasurements">
605   <xs:sequence>
606     <xs:element name="measurementSiteReference" type="D2LogicalModel:_
607 ↪ MeasurementSiteRecordVersionedReference" minOccurs="1" maxOccurs="1" />
608     <xs:element name="measurementTimeDefault" type="D2LogicalModel:DateTime"
609 ↪ minOccurs="1" maxOccurs="1" />
610     <xs:element name="measuredValue" type="D2LogicalModel:_
611 ↪ SiteMeasurementsIndexMeasuredValue" minOccurs="0" maxOccurs="unbounded" />
612     <xs:element name="siteMeasurementsExtension" type="D2LogicalModel:_ExtensionType"
613 ↪ " minOccurs="0" />
614   </xs:sequence>
615 </xs:complexType>
616 <xs:complexType name="SpeedValue">
617   <xs:complexContent>
618     <xs:extension base="D2LogicalModel:DataValue">
619       <xs:sequence>
620         <xs:element name="speed" type="D2LogicalModel:KilometresPerHour" minOccurs=

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606   <xs:element name="speedValueExtension" type="D2LogicalModel:_ExtensionType"
607   </xs:sequence>
608   </xs:extension>
609   </xs:complexContent>
610 </xs:complexType>
611 <xs:simpleType name="String">
612   <xs:restriction base="xs:string">
613     <xs:maxLength value="1024" />
614   </xs:restriction>
615 </xs:simpleType>
616 <xs:complexType name="Temperature">
617   <xs:sequence>
618     <xs:element name="airTemperature" type="D2LogicalModel:TemperatureValue"
619     <xs:element name="dewPointTemperature" type="D2LogicalModel:TemperatureValue"
620     <xs:element name="maximumTemperature" type="D2LogicalModel:TemperatureValue"
621     <xs:element name="minimumTemperature" type="D2LogicalModel:TemperatureValue"
622     <xs:element name="temperatureExtension" type="D2LogicalModel:_ExtensionType"
623   </xs:sequence>
624 </xs:complexType>
625 <xs:simpleType name="TemperatureCelsius">
626   <xs:restriction base="D2LogicalModel:Float" />
627 </xs:simpleType>
628 <xs:complexType name="TemperatureInformation">
629   <xs:complexContent>
630     <xs:extension base="D2LogicalModel:WeatherData">
631       <xs:sequence>
632         <xs:element name="temperature" type="D2LogicalModel:Temperature" />
633         <xs:element name="temperatureInformationExtension" type="D2LogicalModel:_
634         </xs:sequence>
635       </xs:extension>
636     </xs:complexContent>
637 </xs:complexType>
638 <xs:complexType name="TemperatureValue">
639   <xs:complexContent>
640     <xs:extension base="D2LogicalModel:DataValue">
641       <xs:sequence>
642         <xs:element name="temperature" type="D2LogicalModel:TemperatureCelsius"
643         <xs:element name="temperatureValueExtension" type="D2LogicalModel:_
644         </xs:sequence>
645       </xs:extension>
646     </xs:complexContent>
647 </xs:complexType>
648 <xs:simpleType name="TimePrecisionEnum">
649   <xs:restriction base="xs:string">
650     <xs:enumeration value="tenthsOfSecond" />
651     <xs:enumeration value="second" />

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652     <xs:enumeration value="minute" />
653     <xs:enumeration value="quarterHour" />
654     <xs:enumeration value="halfHour" />
655     <xs:enumeration value="hour" />
656   </xs:restriction>
657 </xs:simpleType>
658 <xs:simpleType name="UrgencyEnum">
659   <xs:restriction base="xs:string">
660     <xs:enumeration value="extremelyUrgent" />
661     <xs:enumeration value="urgent" />
662     <xs:enumeration value="normalUrgency" />
663   </xs:restriction>
664 </xs:simpleType>
665 <xs:complexType name="VehicleCountValue">
666   <xs:complexContent>
667     <xs:extension base="D2LogicalModel:DataValue">
668       <xs:sequence>
669         <xs:element name="vehicleCount" type="D2LogicalModel:NonNegativeInteger"
670 ↪ minOccurs="1" maxOccurs="1" />
671         <xs:element name="vehicleCountValueExtension" type="D2LogicalModel:_
672 ↪ ExtensionType" minOccurs="0" />
673       </xs:sequence>
674     </xs:extension>
675   </xs:complexContent>
676 </xs:complexType>
677 <xs:complexType name="VersionedReference">
678   <xs:attribute name="id" type="xs:string" use="required" />
679   <xs:attribute name="version" type="xs:string" use="required" />
680 </xs:complexType>
681 <xs:complexType name="Visibility">
682   <xs:sequence>
683     <xs:element name="minimumVisibilityDistance" type=
684 ↪ "D2LogicalModel:IntegerMetreDistanceValue" />
685     <xs:element name="visibilityExtension" type="D2LogicalModel:_ExtensionType"
686 ↪ minOccurs="0" />
687   </xs:sequence>
688 </xs:complexType>
689 <xs:complexType name="VisibilityInformation">
690   <xs:complexContent>
691     <xs:extension base="D2LogicalModel:WeatherData">
692       <xs:sequence>
693         <xs:element name="visibility" type="D2LogicalModel:Visibility" />
694         <xs:element name="visibilityInformationExtension" type="D2LogicalModel:_
695 ↪ ExtensionType" minOccurs="0" />
696       </xs:sequence>
697     </xs:extension>
698   </xs:complexContent>
699 </xs:complexType>
700 <xs:complexType name="VmsFault">
701   <xs:complexContent>
702     <xs:extension base="D2LogicalModel:Fault">
703       <xs:sequence>
704         <xs:element name="vmsFault" type="D2LogicalModel:VmsFaultEnum" minOccurs="1
705 ↪ " maxOccurs="1" />
706         <xs:element name="vmsFaultExtension" type="D2LogicalModel:_ExtensionType"
707 ↪ minOccurs="0" />

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701     </xs:sequence>
702   </xs:extension>
703 </xs:complexContent>
704 </xs:complexType>
705 <xs:simpleType name="VmsFaultEnum">
706   <xs:restriction base="xs:string">
707     <xs:enumeration value="communicationsFailure" />
708     <xs:enumeration value="incorrectMessageDisplayed" />
709     <xs:enumeration value="incorrectPictogramDisplayed" />
710     <xs:enumeration value="outOfService" />
711     <xs:enumeration value="powerFailure" />
712     <xs:enumeration value="unableToClearDown" />
713     <xs:enumeration value="unknown" />
714     <xs:enumeration value="other" />
715   </xs:restriction>
716 </xs:simpleType>
717 <xs:complexType name="VmsUnitFault">
718   <xs:complexContent>
719     <xs:extension base="D2LogicalModel:Fault">
720       <xs:sequence>
721         <xs:element name="vmsUnitFault" type="D2LogicalModel:VmsFaultEnum"
722 ↪ minOccurs="1" maxOccurs="1" />
723         <xs:element name="vmsUnitFaultExtension" type="D2LogicalModel:_ExtensionType"
724 ↪ minOccurs="0" />
725       </xs:sequence>
726     </xs:extension>
727   </xs:complexContent>
728 </xs:complexType>
729 <xs:complexType name="WeatherData" abstract="true">
730   <xs:complexContent>
731     <xs:extension base="D2LogicalModel:BasicData">
732       <xs:sequence>
733         <xs:element name="weatherDataExtension" type="D2LogicalModel:_ExtensionType"
734 ↪ minOccurs="0" />
735       </xs:sequence>
736     </xs:extension>
737   </xs:complexContent>
738 </xs:complexType>
739 <xs:simpleType name="WeatherRelatedRoadConditionTypeEnum">
740   <xs:restriction base="xs:string">
741     <xs:enumeration value="blackIce" />
742     <xs:enumeration value="deepSnow" />
743     <xs:enumeration value="dry" />
744     <xs:enumeration value="freezingOfWetRoads" />
745     <xs:enumeration value="freezingPavements" />
746     <xs:enumeration value="freezingRain" />
747     <xs:enumeration value="freshSnow" />
748     <xs:enumeration value="ice" />
749     <xs:enumeration value="iceBuildUp" />
750     <xs:enumeration value="iceWithWheelBarTracks" />
751     <xs:enumeration value="icyPatches" />
752     <xs:enumeration value="looseSnow" />
753     <xs:enumeration value="normalWinterConditionsForPedestrians" />
754     <xs:enumeration value="packedSnow" />
755     <xs:enumeration value="roadSurfaceMelting" />
756     <xs:enumeration value="slipperyRoad" />

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754     <xs:enumeration value="slushOnRoad" />
755     <xs:enumeration value="slushStrings" />
756     <xs:enumeration value="snowDrifts" />
757     <xs:enumeration value="snowOnPavement" />
758     <xs:enumeration value="snowOnTheRoad" />
759     <xs:enumeration value="surfaceWater" />
760     <xs:enumeration value="wet" />
761     <xs:enumeration value="wetAndIcyRoad" />
762     <xs:enumeration value="wetIcyPavement" />
763     <xs:enumeration value="other" />
764   </xs:restriction>
765 </xs:simpleType>
766 <xs:complexType name="Wind">
767   <xs:sequence>
768     <xs:element name="windMeasurementHeight" type=
769 ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
770     <xs:element name="windSpeed" type="D2LogicalModel:SpeedValue" minOccurs="0" />
771     <xs:element name="maximumWindSpeed" type="D2LogicalModel:SpeedValue" minOccurs=
772 ↪ "0" />
773     <xs:element name="windDirectionBearing" type=
774 ↪ "D2LogicalModel:DirectionBearingValue" minOccurs="0" />
775     <xs:element name="windDirectionCompass" type=
776 ↪ "D2LogicalModel:DirectionCompassValue" minOccurs="0" />
777     <xs:element name="windExtension" type="D2LogicalModel:_ExtensionType" minOccurs=
778 ↪ "0" />
779   </xs:sequence>
780 </xs:complexType>
781 <xs:complexType name="WindInformation">
782   <xs:complexContent>
783     <xs:extension base="D2LogicalModel:WeatherData">
784       <xs:sequence>
785         <xs:element name="wind" type="D2LogicalModel:Wind" />
786         <xs:element name="windInformationExtension" type="D2LogicalModel:_
787 ↪ ExtensionType" minOccurs="0" />
788       </xs:sequence>
789     </xs:extension>
790   </xs:complexContent>
791 </xs:complexType>
792 </xs:schema>

```


CHANGELOG

version 1.0.0

- Initial format and documentation

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